

BRENT LEE JONES

Arkansas, US

PROFILE SUMMARY

Principal Software Engineer with 18+ years of experience architecting and delivering production software across frontend, backend, cloud infrastructure, distributed systems, and AI-powered applications. Deep hands-on expertise across React, TypeScript, Node.js, Python, PHP/Laravel, SQL and NoSQL databases, APIs, Kubernetes, and cloud-native architecture, with a career spanning enterprise systems, large-scale SaaS platforms, and modern AI products. Proven ability to move across system boundaries-from domain modeling and application architecture to production implementation, performance, reliability, and infrastructure-while leading technical decisions that keep complex systems scalable and maintainable. Combines principal-level architectural judgment with hands-on engineering, working closely with product and engineering teams to turn complex business requirements into reliable software built to evolve.

CORE COMPETENCIES

- Extensive expertise architecting and delivering large-scale SaaS and enterprise software across frontend, backend, data, cloud infrastructure, and distributed systems, with a focus on scalable architecture, clear system boundaries, reliability, and long-term maintainability.
- Deep hands-on experience with modern full-stack development using React, Next.js, TypeScript, JavaScript, Node.js, Python, FastAPI, PHP, Laravel, and related frameworks, designing production applications from user-facing workflows through backend services and infrastructure.
- Strong background in backend and API architecture, including REST, GraphQL, microservices, event-driven systems, asynchronous processing, third-party integrations, authentication, authorization, and complex business workflows across distributed applications.
- Advanced experience designing data architectures across PostgreSQL, MySQL, SQL Server, MongoDB, Supabase, Redis, and other SQL/NoSQL technologies, including domain modeling, schema design, query optimization, indexing, data integrity, caching, and high-volume transactional workloads.
- Extensive cloud and platform engineering experience with AWS, containerized applications, Docker, Kubernetes, CI/CD, serverless and cloud-native services, with a strong focus on production reliability, scalability, observability, security, and repeatable deployment.
- Deep expertise building AI-powered SaaS products and integrating LLM capabilities into production systems, including OpenAI, Anthropic Claude, prompt engineering, structured outputs, tool/function calling, embeddings, vector search, RAG, and multi-model AI architectures.
- Experienced in designing AI agents and agentic workflows that combine LLM reasoning with application tools, APIs, databases, retrieval systems, business rules, and human-in-the-loop controls while preserving deterministic behavior where production systems require it.
- Strong understanding of production AI architecture including RAG pipelines, document ingestion and chunking, embedding strategies, vector databases, context management, model routing, evaluation, guardrails, observability, latency and cost optimization, and reliable integration of AI capabilities with existing SaaS workflows.
- Proven ability to design secure multi-tenant SaaS architectures incorporating authentication, role-based access control, tenant isolation, API security, data ownership, background processing, real-time workflows, third-party services, and operational administration.
- Principal-level technical leadership across system design, architecture reviews, domain modeling, performance optimization, production debugging, platform evolution, and engineering standards, combining architectural judgment with hands-on implementation while working across product, engineering, and business boundaries.

AREAS OF EXPERTISE

Frontend: React | Next.js | TypeScript | JavaScript (ES6+) | HTML5 | CSS3 | Material UI (MUI) | Responsive Web Applications

Backend: Node.js | Python | FastAPI | PHP | Laravel | CodeIgniter | C# | REST APIs | GraphQL | SOAP | Microservices | API Integration | Distributed Systems | Event-Driven Architecture

Databases & Data: PostgreSQL | MySQL | MongoDB | SQL Server | Supabase | Redis | SQL | NoSQL | Data Modeling | Schema Design | Query Optimization | Caching

AI & LLM Engineering: OpenAI API | Anthropic Claude | LLMs | Generative AI | AI Agents | Agentic Workflows | Retrieval-Augmented Generation (RAG) | Embeddings | Vector Search | Vector Databases | Prompt Engineering | Tool Calling | Function Calling | Structured Outputs | LLM Evaluation | AI Guardrails | Model Routing | Context Management

Cloud & DevOps: AWS | Docker | Kubernetes | CI/CD | Cloud-Native Architecture | Serverless Architecture | Containerization | Infrastructure Automation | Production Deployment | Observability | Monitoring | Scalability | Reliability

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Architecture & Engineering: SaaS Architecture | Multi-Tenant Architecture | System Design | Software Architecture | Domain Modeling | Clean Architecture | API Architecture | Distributed Architecture | Authentication & Authorization | Role-Based Access Control (RBAC) | Performance Optimization | Production Debugging | Security | Technical Leadership

PROFESSIONAL EXPERIENCE

SELF-EMPLOYED

PRINCIPAL AI & SAAS ENGINEER

MAR 2023 – PRESENT

- ARCHITECT AND DELIVER PRODUCTION AI-POWERED SAAS PLATFORMS ACROSS FRONTEND, BACKEND, DATA, CLOUD INFRASTRUCTURE, AND AI SERVICES USING NEXT.JS, REACT, TYPESCRIPT, PYTHON, FASTAPI, SUPABASE, AND POSTGRESQL.
- DESIGN LLM-POWERED SYSTEMS INTEGRATING GENERATIVE AI, RETRIEVAL-AUGMENTED GENERATION (RAG), AI AGENTS, TOOL CALLING, STRUCTURED OUTPUTS, RETRIEVAL PIPELINES, AND EXTERNAL APIS INTO OPERATIONAL BUSINESS WORKFLOWS.
- LEAD END-TO-END SYSTEM ARCHITECTURE INCLUDING DOMAIN MODELING, API DESIGN, MULTI-TENANT DATA ARCHITECTURE, AUTHENTICATION AND AUTHORIZATION, CLOUD DEPLOYMENT, THIRD-PARTY INTEGRATIONS, AND PRODUCTION RELIABILITY.
- ESTABLISH ARCHITECTURAL BOUNDARIES BETWEEN DETERMINISTIC BUSINESS LOGIC AND AI CAPABILITIES, ALLOWING LLM-DRIVEN FUNCTIONALITY TO EVOLVE WITHOUT COMPROMISING CORE APPLICATION BEHAVIOR, SECURITY, OR MAINTAINABILITY.
- OWN TECHNICAL STRATEGY FROM SYSTEM DESIGN THROUGH HANDS-ON IMPLEMENTATION, PRODUCTION DEPLOYMENT, PERFORMANCE OPTIMIZATION, AND ARCHITECTURAL EVOLUTION FOR STARTUP AND ENTERPRISE SAAS PRODUCTS.

ACTITRAK

STAFF SOFTWARE ENGINEER

JUN 2019 – FEB 2023

- ARCHITECTED CORE SAAS PLATFORM SERVICES SUPPORTING MILLIONS OF DAILY EVENTS ACROSS DISTRIBUTED CLOUD INFRASTRUCTURE, WITH EMPHASIS ON SCALABILITY, RELIABILITY, PERFORMANCE, AND OPERATIONAL RESILIENCE.
- LED PLATFORM ENGINEERING INITIATIVES SPANNING DISTRIBUTED SYSTEMS, BACKEND SERVICES, APIS, DATA-INTENSIVE WORKFLOWS, AND CLOUD-NATIVE ARCHITECTURE ACROSS MULTIPLE ENGINEERING TEAMS.
- DROVE ARCHITECTURE AND IMPLEMENTATION OF AI-POWERED PRODUCT CAPABILITIES, INTEGRATING INTELLIGENT FUNCTIONALITY INTO ESTABLISHED PRODUCTION WORKFLOWS AND PLATFORM SERVICES.
- LED SYSTEM DESIGN REVIEWS, PERFORMANCE AND SCALABILITY INITIATIVES, PRODUCTION PROBLEM SOLVING, AND ARCHITECTURAL DECISIONS AFFECTING SHARED PLATFORM CAPABILITIES.
- PROVIDED CROSS-TEAM TECHNICAL LEADERSHIP, ESTABLISHING ENGINEERING PATTERNS AND SYSTEM BOUNDARIES THAT IMPROVED MAINTAINABILITY, DEVELOPMENT VELOCITY, AND LONG-TERM PLATFORM EVOLUTION.

FIRST ORION

LEAD SOFTWARE ENGINEER

APR 20215 – MAY 2019

- LED THE ARCHITECTURE AND DELIVERY OF LARGE-SCALE SAAS PLATFORMS USING MICROSERVICES, CLOUD-NATIVE SERVICES, APIS, AND DISTRIBUTED APPLICATION PATTERNS.
- DESIGNED SERVICE BOUNDARIES, BACKEND ARCHITECTURES, INTEGRATION PATTERNS, AND DATA FLOWS SUPPORTING SCALABLE BUSINESS CAPABILITIES ACROSS PRODUCTION SYSTEMS.
- LED ENGINEERS THROUGH ARCHITECTURE DECISIONS, IMPLEMENTATION STRATEGY, CODE REVIEWS, PRODUCTION TROUBLESHOOTING, AND THE EVOLUTION OF SHARED PLATFORM COMPONENTS.
- WORKED CLOSELY WITH PRODUCT AND ENGINEERING TEAMS TO TRANSLATE COMPLEX BUSINESS REQUIREMENTS INTO MAINTAINABLE TECHNICAL ARCHITECTURES AND PRODUCTION-READY CAPABILITIES.
- IMPROVED PLATFORM SCALABILITY, RELIABILITY, AND ENGINEERING VELOCITY BY REDUCING COUPLING BETWEEN SERVICES AND ESTABLISHING REUSABLE ARCHITECTURAL PATTERNS ACROSS THE SYSTEM.

ELYXOR

SENIOR SOFTWARE ENGINEER

JUL 2011 – MAR 2015

- DESIGNED AND DELIVERED DISTRIBUTED BACKEND SYSTEMS, ENTERPRISE WEB APPLICATIONS, CLOUD-BASED SERVICES, AND API-DRIVEN INTEGRATIONS FOR ENTERPRISE CLIENTS.
- LED REST API AND SERVICE ARCHITECTURE, ESTABLISHING REUSABLE INTEGRATION PATTERNS AND MAINTAINABLE BOUNDARIES

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BETWEEN APPLICATIONS, BUSINESS SERVICES, AND EXTERNAL SYSTEMS.

- DESIGNED BACKEND COMPONENTS AND DATA-ACCESS LAYERS WITH EMPHASIS ON PERFORMANCE, SCALABILITY, SECURITY, AND MAINTAINABILITY ACROSS ENTERPRISE WORKLOADS.
- DIAGNOSED AND RESOLVED COMPLEX APPLICATION, API, DATABASE, AND PERFORMANCE ISSUES ACROSS MULTI-TIER PRODUCTION ENVIRONMENTS.
- MENTORED ENGINEERS, CONDUCTED CODE AND DESIGN REVIEWS, AND HELPED ESTABLISH ENGINEERING PRACTICES FOR RELIABLE DELIVERY OF ENTERPRISE SOFTWARE.

MAINSTREAM TECHNOLOGIES

SOFTWARE ENGINEER

JUL 2007 – JUN 2011

- DEVELOPED ENTERPRISE WEB APPLICATIONS AND BACKEND SERVICES USING C#, SQL SERVER, REST/SOAP APIS, AND SERVICE-ORIENTED APPLICATION ARCHITECTURES.
- DESIGNED REUSABLE APPLICATION COMPONENTS, BACKEND BUSINESS LOGIC, DATABASE INTEGRATIONS, AND SERVICE INTERFACES SUPPORTING COMPLEX ENTERPRISE WORKFLOWS.
- OPTIMIZED SQL QUERIES, DATA-ACCESS PATTERNS, AND APPLICATION PERFORMANCE WHILE MAINTAINING DATA INTEGRITY ACROSS PRODUCTION BUSINESS SYSTEMS.
- INTEGRATED INTERNAL AND THIRD-PARTY SERVICES THROUGH REST AND SOAP APIS, SUPPORTING RELIABLE COMMUNICATION ACROSS ENTERPRISE APPLICATIONS.
- COLLABORATED WITH ENGINEERING, PRODUCT, AND BUSINESS TEAMS TO TRANSLATE OPERATIONAL REQUIREMENTS INTO MAINTAINABLE, PRODUCTION-READY SOFTWARE.

EDUCATION

UNIVERSITY OF ARKANSAS AT FAYETTEVILLE: BACHELOR OF COMPUTER SCIENCE 2001-2005

TEXAS A&M UNIVERSITY: MASTER OF COMPUTER SCIENCE 2005-2007